* OUTPUT

Enter host name or IP address. google.com

Pinging google.com

Pinging 142.250.195.14

Reply from ~~goog~~ google.com : time = 0ms

Reply from google.com : time = 0ms

Reply from google.com : time = 0ms

Packets: sent = 4, Received = 4, Lost = 0

```
root@ip-10-48-165-144:~/Documents# # UDP Ping
root@ip-10-48-165-144:~/Documents# sudo nmap -PU -sn 10.48.165.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-12 15:41 UTC
Nmap scan report for ip-10-48-165-61.ap-south-1.compute.internal (10.48.165.61)
Host is up (0.00010s latency).
MAC Address: 02:FF:F8:31:E2:75 (Unknown)
Nmap scan report for ip-10-48-165-93.ap-south-1.compute.internal (10.48.165.93)
Host is up (0.000032s latency).
MAC Address: 02:FF:CD:46:17:79 (Unknown)
Nmap scan report for ip-10-48-165-97.ap-south-1.compute.internal (10.48.165.97)
Host is up (0.000072s latency).
MAC Address: 02:FF:EA:89:DB:47 (Unknown)
Nmap scan report for ip-10-48-165-105.ap-south-1.compute.internal (10.48.165.105)
Host is up (0.000036s latency).
MAC Address: 02:FF:E9:2F:DF:B3 (Unknown)
Nmap scan report for ip-10-48-165-150.ap-south-1.compute.internal (10.48.165.150)
Host is up (0.000051s latency).
MAC Address: 02:FF:D7:B6:13:3B (Unknown)
Nmap scan report for ip-10-48-165-200.ap-south-1.compute.internal (10.48.165.200)
Host is up (0.00011s latency).
MAC Address: 02:FF:DD:F3:44:2D (Unknown)
Nmap scan report for ip-10-48-165-215.ap-south-1.compute.internal (10.48.165.215)
Host is up (0.000061s latency).
MAC Address: 02:FF:D0:81:42:2B (Unknown)
Nmap scan report for ip-10-48-165-237.ap-south-1.compute.internal (10.48.165.237)
Host is up (0.000044s latency).
MAC Address: 02:FF:ED:D6:B5:39 (Unknown)
Nmap scan report for ip-10-48-165-254.ap-south-1.compute.internal (10.48.165.254)
Host is up (0.000040s latency).
MAC Address: 02:FF:C2:68:84:A5 (Unknown)
Nmap scan report for ip-10-48-165-144.ap-south-1.compute.internal (10.48.165.144)
Host is up.
Nmap done: 256 IP addresses (10 hosts up) scanned in 2.50 seconds
root@ip-10-48-165-144:~/Documents#
```

Exp no: 13

Date: 7/4/19

IMPLEMENT YOUR OWN PING PROGRAM

* Aim: To check host reachability and measure response time using a ping program.

* Note: Java's standard InetAddress.isReachable() is a portable reachability check, but the operating system may implement it using ICMP or another mechanism. This program is suitable for a simple lab demonstration. It is not a custom raw-ICMP implementation.

* Ping Program.java

```
import java.net.*;
import java.util.Scanner;

public class PingProgram {
    public static void main(String[] args) {
        try(Scanner input = new Scanner(System.in)){
        System.out.print("Enter host name or IP address: ");
        String host = input.nextLine().trim();
        InetAddress address = InetAddress.getByName(host);
        System.out.println("Pinging " + address.getHostAddress());
        int count=4;
        int received=0;
```

```
root@ip-10-48-165-144:~/Documents# # TCP SYN Ping (default port 80)
root@ip-10-48-165-144:~/Documents# sudo nmap -PS -sn 10.48.165.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-12 15:41 UTC
Nmap scan report for ip-10-48-165-61.ap-south-1.compute.internal (10.48.165.61)
Host is up (0.000072s latency).
MAC Address: 02:FF:F8:31:E2:75 (Unknown)
Nmap scan report for ip-10-48-165-93.ap-south-1.compute.internal (10.48.165.93)
Host is up (0.000043s latency).
MAC Address: 02:FF:CD:46:17:79 (Unknown)
Nmap scan report for ip-10-48-165-97.ap-south-1.compute.internal (10.48.165.97)
Host is up (0.000027s latency).
MAC Address: 02:FF:EA:89:DB:47 (Unknown)
Nmap scan report for ip-10-48-165-105.ap-south-1.compute.internal (10.48.165.105)
Host is up (0.000051s latency).
MAC Address: 02:FF:E9:2E:DF:B3 (Unknown)
Nmap scan report for ip-10-48-165-150.ap-south-1.compute.internal (10.48.165.150)
Host is up (0.000077s latency).
MAC Address: 02:FF:D7:B6:13:3B (Unknown)
Nmap scan report for ip-10-48-165-200.ap-south-1.compute.internal (10.48.165.200)
Host is up (0.000029s latency).
MAC Address: 02:FF:DD:F3:44:2D (Unknown)
Nmap scan report for ip-10-48-165-215.ap-south-1.compute.internal (10.48.165.215)
Host is up (0.000075s latency).
MAC Address: 02:FF:D0:81:42:2B (Unknown)
Nmap scan report for ip-10-48-165-237.ap-south-1.compute.internal (10.48.165.237)
Host is up (0.000058s latency).
MAC Address: 02:FF:ED:D6:B5:39 (Unknown)
Nmap scan report for ip-10-48-165-254.ap-south-1.compute.internal (10.48.165.254)
Host is up (0.000038s latency).
MAC Address: 02:FF:C2:68:84:A5 (Unknown)
Nmap scan report for ip-10-48-165-144.ap-south-1.compute.internal (10.48.165.144)
Host is up.
Nmap done: 256 IP addresses (10 hosts up) scanned in 2.49 seconds
root@ip-10-48-165-144:~/Documents#
```

```
root@ip-10-48-165-144:~/Documents# # ICMP Echo discovery
root@ip-10-48-165-144:~/Documents# sudo nmap -PE -sn 10.48.165.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-12 15:38 UTC
Nmap scan report for ip-10-48-165-61.ap-south-1.compute.internal (10.48.165.61)
Host is up (0.000041s latency).
MAC Address: 02:FF:F8:31:E2:75 (Unknown)
Nmap scan report for ip-10-48-165-93.ap-south-1.compute.internal (10.48.165.93)
Host is up (0.000057s latency).
MAC Address: 02:FF:CD:46:17:79 (Unknown)
Nmap scan report for ip-10-48-165-97.ap-south-1.compute.internal (10.48.165.97)
Host is up (0.000040s latency).
MAC Address: 02:FF:EA:89:DB:47 (Unknown)
Nmap scan report for ip-10-48-165-105.ap-south-1.compute.internal (10.48.165.105)
Host is up (0.000041s latency).
MAC Address: 02:FF:E9:2E:DF:B3 (Unknown)
Nmap scan report for ip-10-48-165-150.ap-south-1.compute.internal (10.48.165.150)
Host is up (0.000035s latency).
MAC Address: 02:FF:D7:B6:13:3B (Unknown)
Nmap scan report for ip-10-48-165-200.ap-south-1.compute.internal (10.48.165.200)
Host is up (0.000041s latency).
MAC Address: 02:FF:DD:F3:44:2D (Unknown)
Nmap scan report for ip-10-48-165-215.ap-south-1.compute.internal (10.48.165.215)
Host is up (0.000054s latency).
MAC Address: 02:FF:D0:81:42:2B (Unknown)
Nmap scan report for ip-10-48-165-237.ap-south-1.compute.internal (10.48.165.237)
Host is up (0.000028s latency).
MAC Address: 02:FF:ED:D6:B5:39 (Unknown)
Nmap scan report for ip-10-48-165-254.ap-south-1.compute.internal (10.48.165.254)
Host is up (0.000065s latency).
MAC Address: 02:FF:C2:68:84:A5 (Unknown)
Nmap scan report for ip-10-48-165-144.ap-south-1.compute.internal (10.48.165.144)
Host is up.
Nmap done: 256 IP addresses (10 hosts up) scanned in 3.71 seconds
root@ip-10-48-165-144:~/Documents#
```

```
for (int i=1; i<=count; i++){
    long start = System.nanoTime();
    boolean reachable = address.isReachable(3000);
    long elapsedMs = (System.nanoTime() - start) / 1_000_000;
    if (reachable){
        received++;
        System.out.println("Reply from " + address.getHostAddress()
                + ": time=" + elapsedMs + "ms");
    } else {
        System.out.println("Request timed out");
    }
} catch (Exception e) {
    System.out.println("Ping Failed: " + e.getMessage());
}
```

* Result:

A basic host-reachability and response-time checking program was implemented.